

IP addressing plan

The client was allocated the network 192.168.61.0/24 by the ISP. To meet the requirements for five departments and ensure efficient use of addresses, Variable Length Subnet Masking (VLSM) was applied. The three large departments (Sales, Repairs, and Admin) were each allocated a /26 subnet providing 62 usable hosts, while the smaller Shared Printer and Future Expansion departments were allocated /27 subnets providing 30 usable hosts each. This design complies with CR8 by sharing the printer and server resources in VLAN 50, and reserves Fa0/9 for future growth. The subnets were allocated from largest to smallest to prevent wastage.

VLAN	Dept	Needed hosts	Subnet to be used	Gateway	Usable hosts
10	sales	60	192.168.61.0/26	192.168.61.1	.2-.62
20	repairs	60	192.168.61.64/26	192.168.61.65	.66-.126
30	admin	30	192.168.61.128/26	192.168.61.129	.130-.190
40	future	30	192.168.61.192/27	192.168.61.193	.194-.222 RESERVED
50	Shared printer	30	192.168.61.224/27	192.168.61.225	.226-.254(server.230 printer.231)